Docker Generation & Testing Manual

Container Architecture, Health Monitoring & Operational Verification

DevOps Manual | Phase 7 Release | Date: 2026-10-23 | Status: Production-Ready

1. Container Architecture & Security Specs

• Base Image: python:3.10-slim (Minimal Debian footprint)
• System Packages: libgl1, libglib2.0-0, curl
• Security User: Non-privileged appuser (UID 1000) for unprivileged execution
• Service Port: 8501 (Mapped 1:1 on host)
• Health Check: curl -f http://localhost:8501/_stcore/health

2. Step-by-Step Container Deployment

1. Ensure Docker Desktop or Docker Engine daemon is running.

2. Navigate to project root:
   cd C:\Pankaja\AUTOMATED_SDLC_AGENT_PROJECTS\lane_detection

3. Build and launch container in detached mode:
   docker compose up --build -d

4. Verify container running state:
   docker ps

5. Inspect live container logs:
   docker logs -f lane_detection_dashboard

6. Access web dashboard in browser: http://localhost:8501

3. Health Monitoring & Troubleshooting FAQ

Healthcheck Verification:
Docker executes automated probes every 30s. The status in 'docker ps' displays '(healthy)' after 20s initial boot.

Troubleshooting Common Issues:
• Port 8501 In Use: If another container occupies port 8501, run 'docker ps' to find its name and 'docker stop <name>'.
• GPU not detected: The container runs high-performance CPU inference (<50ms per frame) and

does not require CUDA drivers.
• Decompression Bombs: Pillow pixel limit is capped at 50,000,000 pixels inside src/utils.py to reject oversized uploads.

4. Container Teardown & Maintenance

To safely stop and remove the container:
docker compose down

To clean unused build cache:
docker system prune -f